

Question 1 (14 marks)

(a) Solve $\log_e x + \log_e (2 - x) = 0$ (3 marks)

(b) The arc length of a sector of a circle is 6 cm. If the circle has a radius of 3 cm, find the area of the sector. (2 marks)

(c) Find the size of the acute angle between the lines whose equations are

$$y = 2\sqrt{3}x - \sqrt{6} \quad \text{and}$$

$$7y = \sqrt{3}x + \sqrt{2} \quad (3 \text{ marks})$$

(d) (i) Write down the expansion of $\cos(A + B)$. (1 mark)

(ii) If $\sin A = \frac{2}{3}$ and $\frac{\pi}{2} < A < \pi$ and $\cos B = \frac{3}{4}$ and $0 < B < \frac{\pi}{2}$
Find the exact value of $\cos(A + B)$ (3 marks)

(e) Find $\frac{dy}{dx}$ if $y = e^x \sin x$ (2 marks)

Question 2 (14 marks) (START QUESTION ON A NEW PAGE)

(a) (i) Show that $\frac{\sin 2x}{\tan x} = 2 \cos^2 x$ (3 marks)

(ii) Hence find $\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan x}$ (1 mark)

(b) Given that $\frac{1}{x+1} - \frac{1}{x+2} = \frac{1}{(x+1)(x+2)}$

Show that $\int_1^3 \frac{dx}{(x+1)(x+2)} = \log_e \frac{6}{5}$ (3 marks)

(c) The gradient at any point (x, y) on a curve is given by $\frac{dy}{dx} = 1 - \frac{6}{x}$. Find the equation of the curve if it passes through the point $(1, 2)$. (2 marks)

Question 2 continued.

(d) (i) Find $\frac{d}{dx}(x \sin x + \cos x)$ **(2 marks)**

(ii) Hence find the value of $\int_{\frac{\pi}{2}}^{\pi} x \cos x dx$ **(3 marks)**

Question 3 (14 marks) (START QUESTION ON A NEW PAGE)

(a) P $(2at, at^2)$ is a point on the parabola $x^2 = 4ay$

(i) Show that the equation of the tangent to this parabola at P is $y = tx - at^2$ **(3 marks)**

(ii) If this tangent meets the x axis at T, find the co-ordinates of T. **(1 mark)**

(iii) M is the mid-point of PT. Find the co-ordinates of M. **(1 mark)**

(iv) Find the equation of the locus of the point M. **(2 marks)**

(b) (i) Write down an expansion for $\cos 2x$ in terms of $\cos^2 x$ **(1 mark)**

(ii) Hence find a primitive function of $\frac{\cos 2x}{\cos^2 x}$ **(2 marks)**

(c) Find $\int \frac{x}{1-x^2} dx$ **(2 marks)**

(d) By using the substitution $u = \sqrt{x}$ or otherwise, find $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ **(2 marks)**

END OF PAPER

Solution

Q1 (a) $\log_e x(2-x) = 0$ (1)

(14) $\therefore x(2-x) = 1$ (1)

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = 1 \quad (1)$$

(b) $6 = 3\theta$

$$\therefore \theta = 2 \text{ radians} \quad (1)$$

$$\therefore A = \frac{1}{2} \times 3^2 \times 2 = 9 \text{ cm}^2 \quad (1)$$

(c) $\left. \begin{array}{l} m_1 = 2\sqrt{3} \\ m_2 = \frac{\sqrt{3}}{7} \end{array} \right\} \quad (1)$

$$\therefore \tan \theta = \frac{2\sqrt{3} - \frac{\sqrt{3}}{7}}{1 + 2\sqrt{3} \cdot \frac{\sqrt{3}}{7}} \quad (1)$$

$$= \frac{14\sqrt{3} - \sqrt{3}}{7 + 6}$$

$$= \frac{13\sqrt{3}}{13}$$

$$= \sqrt{3}$$

$$\therefore \theta = 60^\circ \quad (1)$$

(d) (i) $\cos A \cdot \cos B - \sin A \cdot \sin B$ (1)

(ii) If $\sin A = \frac{2}{3}$, $\cos A = -\frac{\sqrt{5}}{3}$ (1)

If $\cos B = \frac{3}{4}$, $\sin B = \frac{\sqrt{7}}{4}$ (1)

$$\therefore \cos(A+B) = -\frac{\sqrt{5}}{3} \cdot \frac{3}{4} - \frac{2}{3} \cdot \frac{\sqrt{7}}{4}$$

$$= -\frac{\sqrt{5}}{4} - \frac{\sqrt{7}}{6} \quad (1)$$

(e) $\frac{dy}{dx} = e^x \cos x + \sin x e^x$ (2)

$$= e^x (\cos x + \sin x)$$

Q2 (a) (i) LHS = $\frac{2 \sin x \cdot \cos x}{\frac{\sin x}{\cos x}}$ (1)

$$= \frac{2 \sin x \cos^2 x}{\sin x} \quad (1)$$

$$= 2 \cos^2 x = \text{RHS} \quad (1)$$

(ii) $\lim_{x \rightarrow 0} \frac{\sin 2x}{\tan x} = \lim_{x \rightarrow 0} 2 \cos^2 x$

$$= 2 \times 1 = 2 \quad (1)$$

$$(b) \int_1^3 \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx = \left[\log_e (x+1) - \log_e (x+2) \right]_1^3 \quad (1)$$

$$= \left[\log_e \frac{(x+1)}{(x+2)} \right]_1^3$$

$$= \log_e \frac{4}{5} - \log_e \frac{2}{3} \quad (1)$$

$$= \log_e \left(\frac{4}{5} \times \frac{3}{2} \right)$$

$$= \log_e \frac{6}{5} \quad (1)$$

$$(c) \quad y = x - 6 \log_e x + c \quad (1)$$

$$2 = 1 - 6 \log_e 1 + c$$

$$2 = 1 - 0 + c$$

$$c = 1 \quad (1)$$

$$\therefore y = x - 6 \log_e x + 1$$

$$(d) (i) \quad \frac{d}{dx} (x \sin x + \cos x) = x \cos x + \sin x \cdot 1 - \sin x \quad (1)$$

$$= x \cos x \quad (1)$$

$$(ii) \quad \therefore \int_{\frac{\pi}{2}}^{\pi} x \cos x dx = \left[x \sin x + \cos x \right]_{\frac{\pi}{2}}^{\pi} \quad (1)$$

$$= \pi \sin \pi + \cos \pi - \left(\frac{\pi}{2} \sin \frac{\pi}{2} + \cos \frac{\pi}{2} \right) \quad (1)$$

$$= \pi \cdot 0 + (-1) - \frac{\pi}{2} \cdot 1 - 0$$

$$= -1 - \frac{\pi}{2} \quad (1)$$

$$Q3 (i) \quad y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a} \quad (1)$$

$$m_T \text{ at } P = \frac{2at}{2a} = t \quad (1)$$

$$\therefore \text{Equation is } y - at^2 = t(x - 2at)$$

$$y - at^2 = tx - 2at^2 \quad (1)$$

$$y = tx - at^2$$

$$(ii) \text{ at } T, y=0 \quad \therefore tx = at^2$$

$$x = at$$

$$\therefore T \text{ is } (at, 0)$$

(1)

$$(iii) \therefore M \text{ is } \left(\frac{2at+at}{2}, \frac{at^2+0}{2} \right)$$

$$\text{ie } \left(\frac{3at}{2}, \frac{at^2}{2} \right)$$

(1)

(iv) $\therefore$ Parametric Eqs of the locus are

$$x = \frac{3at}{2}, \quad y = \frac{at^2}{2}$$

$$\therefore t = \frac{2x}{3a}$$

(1)

$$\therefore y = a \frac{\frac{4x^2}{9a^2}}{2}$$

$$\text{or } y = \frac{2x^2}{9a}$$

(1)

$$(a) (i) \quad 2 \cos^2 x - 1$$

(1)

$$(ii) \quad \frac{2 \cos^2 x - 1}{\cos^2 x} = 2 - \sec^2 x$$

(1)

$$\therefore \text{A primitive is } 2x - \tan x$$

(1)

$$(c) \quad \int \frac{x}{1-x^2} dx = -\frac{1}{2} \int \frac{-2x}{1-x^2}$$

(1)

$$= -\frac{1}{2} \log_e (1-x^2) + c$$

(1) (no penal for leaving out c)

$$(d) \text{ Let } u = x^{\frac{1}{2}}$$

$$\therefore \frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$2 du = \frac{dx}{\sqrt{x}}$$

(1)

$$\therefore \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx = 2 \int e^u du$$

$$= 2 e^u + c$$

$$= 2 e^{\sqrt{x}} + c$$

(1)

otherwise

indicates

that some well do it by inspection.

no penalty for leaving out c.